

Kafka Topic Creation Failure SOP

Objective

Resolve producer failures caused by missing Kafka topics or unsuccessful topic creation.

Scope

Applicable to Kafka-based applications, streaming platforms, backend services, data engineering pipelines, DevOps teams, and platform teams supporting production or synthetic incident workflows.

Pre-requisites

1. Access to Kafka cluster, namespace or broker host.
2. Access to monitoring dashboards such as Grafana, Prometheus, or cloud monitoring.
3. Kafka CLI tools: kafka-topics, kafka-consumer-groups, kafka-console-producer, and kafka-console-consumer.
4. Permission to view application logs, broker logs, and deployment status.
5. Escalation contact for DevOps, platform, and application owner teams.

Incident Metadata

Field	Value
Ticket ID	T015
Issue	Kafka topic not created
Category	Streaming
Service	Kafka
Severity	Medium
Status	Resolved
Description	Producer failing to send messages
Expected Resolution	Create topic manually or enable auto creation

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Confirm producer error and identify the missing topic name from logs or application configuration.
2. Check whether the topic exists using kafka-topics --list.
3. Verify whether auto topic creation is enabled or disabled for the environment.
4. Create the topic manually with approved partitions, replication factor, and retention policy.
5. Validate ACLs for producer and consumer principals.
6. Retry producer flow with a small test message.
7. Confirm topic appears in monitoring and metadata responses.
8. Update ticket with topic name, configuration, and validation result.

Common Commands

```
kafka-topics.sh --bootstrap-server <broker>:9092 --describe
kafka-consumer-groups.sh --bootstrap-server <broker>:9092 --describe --group <group>
kafka-broker-api-versions.sh --bootstrap-server <broker>:9092
kubectl get pods -n <namespace> | grep kafka
kubectl logs <kafka-pod> -n <namespace> --tail=100
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Topic not found	Topic not created	Create topic with approved config
Producer cannot write	Missing ACL	Grant write permission
Topic created incorrectly	Wrong partition or retention	Update topic config after approval
Auto creation failed	Disabled cluster setting	Use manual creation process

Best Practices

1. Use environment-specific topic naming standards.
2. Create topics through controlled automation where possible.
3. Confirm replication factor and retention before production usage.
4. Avoid enabling auto topic creation in tightly governed production clusters.

References

Apache Kafka Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | Kafka CLI Runbook